

**Definition****Compactness**

A topological space  $(X, T)$  is **compact** if every open cover has a finite subcover. That is, if

$$X = \bigcup_{\alpha \in \Omega} U_{\alpha}$$

is an open cover, then there is some finite subcover

$$X = U_{\alpha_1} \cup U_{\alpha_2} \cup \dots \cup U_{\alpha_n}$$

for some  $\alpha_1, \dots, \alpha_n \in \Omega$ .

**Remark**

Suppose  $Y$  is a subset of a topological space  $(X, T)$ . Then we have notions of  $\text{open}_X$  and  $\text{open}_Y$ . So an open cover of  $Y$  is a collection  $U_{\alpha}$  of  $\text{open}_Y$  sets such that  $Y \subseteq \bigcup U_{\alpha}$ , and a finite subcover is  $Y \subseteq U_{\alpha_1} \cup \dots \cup U_{\alpha_n}$ .

Note here that open covers (and subcovers) may contain more than the subset. However, when restricted to the subspace topology, the union becomes equal to the subset.

**Theorem**

Suppose  $A, B \subseteq X$  are compact subspaces of a topological space  $(X, T)$ . Then their union  $A \cup B$  is also compact.

*Proof:* Let  $A \cup B \subseteq \bigcup U_{\alpha}$  be an open cover of the union. Then it is an open cover of  $A$  since  $A \subseteq A \cup B$ , so it has a finite subcover

$$A \subseteq U_{\alpha_1} \cup \dots \cup U_{\alpha_n}.$$

Similarly,

$$B \subseteq U_{\beta_1} \cup \dots \cup U_{\beta_m}.$$

Then

$$A \cup B \subseteq U_{\alpha_1} \cup \dots \cup U_{\alpha_n} \cup U_{\beta_1} \cup \dots \cup U_{\beta_m}$$

is a finite subcover of the union. So  $A \cup B$  is compact. ■

**Theorem**

Suppose  $(X, T)$  is a compact topological space and  $A \subseteq X$  is closed. Then  $A$  is a compact subset.

*Proof:* Suppose  $A \subseteq \bigcup U_{\alpha}$  is an open cover of  $A$ .

If  $X = \bigcup U_\alpha$ , then this has a finite subcover because  $X$  is compact, meaning

$$A \subseteq U_{\alpha_1} \cup \dots \cup U_{\alpha_n}.$$

If this open cover is a proper subset of  $X$ , then we can add one more open set, namely  $X \setminus A$ .

So  $X = \bigcup U_\alpha \cup (X \setminus A)$  is an open cover of  $X$ , hence it has a finite subcover

$$X = (X \setminus A) \cup U_{\alpha_1} \cup \dots \cup U_{\alpha_n}.$$

Then

$$A \subseteq U_{\alpha_1} \cup \dots \cup U_{\alpha_n}$$

so in either case, this open cover of  $A$  has a finite subcover. Thus,  $A$  is compact. ■

### Theorem

Suppose  $(X, T)$  is a topological space that is Hausdorff, and  $A \subseteq X$  is a compact subset. Then  $A$  is closed.

*Proof:* We prove that  $X \setminus A$  is open.

Let  $x \in X \setminus A$ . Then because  $A$  is Hausdorff, for every  $a \in A$ , we can find disjoint open sets  $U_a$  and  $V_a$  such that  $a \in U_a$  and  $x \in V_a$ .

Now  $A \subseteq \bigcup_{a \in A} U_a$  is an open cover of  $A$ , so it has a finite subcover

$$A \subseteq U_{a_1} \cup \dots \cup U_{a_n}.$$

Given these indices, we know  $x \in V_{a_1} \cap \dots \cap V_{a_n} = V$ . Then  $V$  is an intersection of finitely many open sets, so  $V$  is open. It is also disjoint from  $A$ , since every point  $a \in A$  is an element of some  $U_{a_i}$  which is disjoint from  $V$ .

Now  $x \in V \subseteq X \setminus A$ , so  $x$  is interior to  $X \setminus A$ . Since  $x$  was arbitrary,  $X \setminus A$  is open. ■

### Theorem

Suppose  $(X, T)$  and  $(Y, U)$  are topological spaces, and suppose  $f : X \rightarrow Y$  is a continuous function. Then if  $X$  is compact,  $f(X)$  is a compact subset of  $Y$ .

*Proof:* Suppose  $f(X) \subseteq \bigcup V_\alpha$  is a cover for sets  $V_\alpha$  open in  $Y$ . Then if  $x \in X$ , we have  $f(x) \in f(X)$ , so  $f(x) \in V_\alpha$  for some  $\alpha$ . Hence  $x \in f^{-1}(V_\alpha)$ , meaning

$$X = \bigcup f^{-1}(V_\alpha)$$

is an open cover of  $X$ . Since  $X$  is compact, there is a finite subcover

$$X = f^{-1}(V_{\alpha_1}) \cup \dots \cup f^{-1}(V_{\alpha_n}).$$

Now for any  $f(x) \in f(X)$ , we know  $x \in X$  and thus  $x \in f^{-1}(V_{\alpha_i})$  for some  $i$ , so indeed  $f(x) \in V_{\alpha_i}$ . Thus

$$f(X) \subseteq V_{\alpha_1} \cup \dots \cup V_{\alpha_n}$$

is a finite subcover of  $f(X)$ . Therefore,  $f(X)$  is compact. ■

### Theorem

Suppose  $(X, T)$  is compact,  $(Y, T)$  is Hausdorff, and  $f : X \rightarrow Y$  is a continuous and bijective function. Then  $f$  is a homeomorphism.

*Proof:* We first prove that  $f$  is a **closed map**, meaning if  $A \subseteq X$  is closed <sub>$X$</sub> , then  $f(A)$  is closed <sub>$Y$</sub> .

Suppose  $A \subseteq X$  is closed. Then we know  $A$  is compact, so  $f(A)$  is also compact. Hence  $f(A)$  is closed because it is a compact subset of a Hausdorff space.

Now we prove that  $f$  is an **open map**. If  $U \subseteq X$  is open <sub>$X$</sub> , then  $X \setminus U$  is closed <sub>$X$</sub> . So  $f(X \setminus U)$  is closed <sub>$Y$</sub> , but  $f(U) \subseteq Y \setminus f(X \setminus U)$  because  $f$  is a bijection.

### Exercise

Explain the above statement.

We now prove  $f^{-1} = g : Y \rightarrow X$  is continuous. Since  $f$  is bijective, we know  $g(y)$  equals the unique  $x$  for which  $f(x) = y$ . Moreover, if  $U$  is an open <sub>$X$</sub>  set,  $g^{-1}(U)$  is the preimage of  $U$ , hence  $y \in g^{-1}(U)$  if and only if  $g(y) \in U$ . This implies  $x \in U$ , meaning  $y \in f(U)$ . So  $g^{-1}(U) = f(U)$ , which is open. Hence  $g$  is continuous. ■

### Example

The space  $[0, 1]$  with the standard topology on  $\mathbb{R}$  is compact.

However, there is no continuous bijection between  $[0, 1]$  and  $[0, 1] \times [0, 1]$ . Isn't that interesting?

### Definition

### Connected

A topological space  $(X, T)$  is **not connected** if there are some open sets  $U$  and  $V$  such that

- $U$  and  $V$  are disjoint
- Their union  $U \cup V$  equals  $X$
- $U$  and  $V$  are both nonempty